

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO: 33. Morphological Gradient Technique

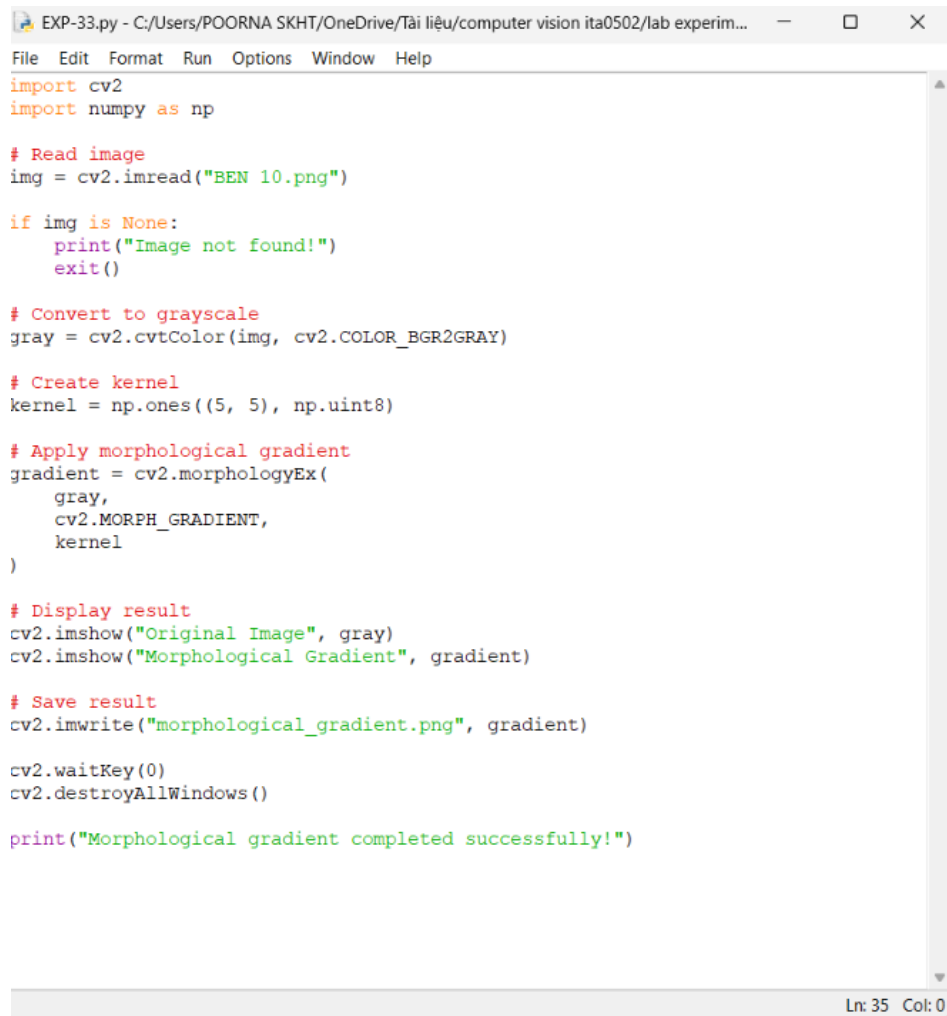
Aim

To perform morphological gradient on an image using OpenCV.

Algorithm

1. Import OpenCV.
2. Read BEN 10.png.
3. Convert the image to grayscale.
4. Create a morphological kernel.
5. Apply morphological gradient using cv2.morphologyEx().
6. Display and save the result.

CODE:

A screenshot of a Python IDE window titled 'EXP-33.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...'. The window contains a Python script for performing a morphological gradient. The code imports cv2 and numpy, reads an image 'BEN 10.png', converts it to grayscale, creates a 5x5 ones kernel, applies cv2.morphologyEx with cv2.MORPH_GRADIENT, displays the original and gradient images, saves the gradient as 'morphological_gradient.png', and prints a success message. The status bar at the bottom right shows 'Ln: 35 Col: 0'.

```
EXP-33.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help

import cv2
import numpy as np

# Read image
img = cv2.imread("BEN 10.png")

if img is None:
    print("Image not found!")
    exit()

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create kernel
kernel = np.ones((5, 5), np.uint8)

# Apply morphological gradient
gradient = cv2.morphologyEx(
    gray,
    cv2.MORPH_GRADIENT,
    kernel
)

# Display result
cv2.imshow("Original Image", gray)
cv2.imshow("Morphological Gradient", gradient)

# Save result
cv2.imwrite("morphological_gradient.png", gradient)

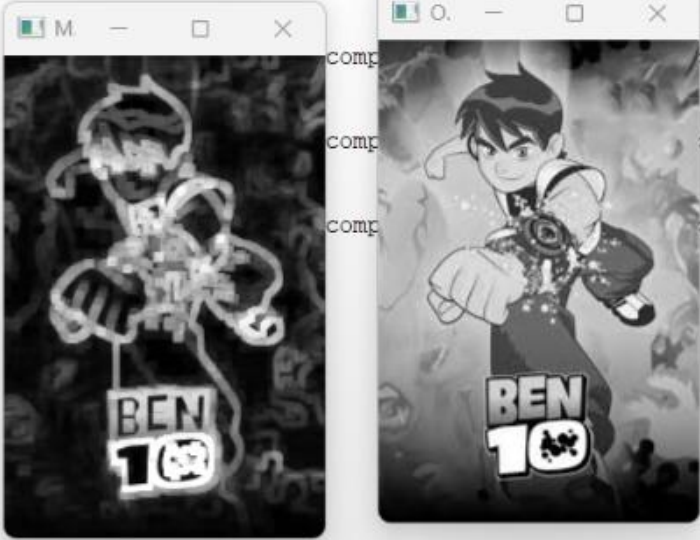
cv2.waitKey(0)
cv2.destroyAllWindows()

print("Morphological gradient completed successfully!")

Ln: 35 Col: 0
```

OUTPUT :

```
*IDLE Shell 3.13.3*
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-26.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-28.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-29.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-30.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-31.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-32.py
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-33.py
```



Result

The morphological gradient is successfully obtained, highlighting the boundaries of objects in the image.